PROGRAMMING IN JAVA

CSA0915

ASSIGNMENT-CO1

CAMPUS SMART-PARKING SYSTEM

TASK 09 – Abstraction and Interfaces

Name: V. Reddy Sreya **Regno: 192525124**

1. Task09.md

Aim

To demonstrate abstraction and interfaces in Java using the Campus Smart-Parking System.

Abstraction

Abstraction hides implementation details and displays only essential features to the user.

Interface

An interface defines methods that implementing classes must provide using the implements keyword.

Application

The ParkingPayment interface defines a common payment method. StudentPayment and VisitorPayment implement their own payment calculations.

Advantages

1. Hides unnecessary implementation details.
2. Improves code reusability.
3. Provides flexibility.
4. Supports multiple implementations.

Conclusion

Abstraction and interfaces help implement different payment methods using a common structure.

2. ParkingPayment.java

```
public interface ParkingPayment {
    double calculatePayment(int hours);
}
```

3. StudentPayment.java

```
public class StudentPayment implements ParkingPayment {
    @Override
    public double calculatePayment(int hours) {
        return hours * 10;
    }
}
```

4. VisitorPayment.java

```
public class VisitorPayment implements ParkingPayment {
    @Override
    public double calculatePayment(int hours) {
        return hours * 30;
    }
}
```

5. Main.java

```
public class Main {
    public static void main(String[] args) {
        int hours = 3;
        ParkingPayment student = new StudentPayment();
        ParkingPayment visitor = new VisitorPayment();
        System.out.println("Campus Parking Payment Details:");
        System.out.println("Parking Duration: " + hours + " hours");
        System.out.println(
            "Student Payment: Rs. " +
            student.calculatePayment(hours)
```

```
        );
        System.out.println(
            "Visitor Payment: Rs. " +
            visitor.calculatePayment(hours)
        );
    }
}
```

6. Output:

